SIH 2026 Proposed Solution & Architecture Report

AEGIS Sovereign Intelligence Report | Generated: 2026-09-02 15:23:38 UTC | Classification: **CONFIDENTIAL INTERNAL**

EXECUTIVE SUMMARY

Aegis proposes a sovereign on-premise AI workbench utilizing open-weight multimodal LLMs for confidential industrial work. The solution focuses on 100% local AI processing, intelligent model routing, agentic workflows (Plan → Execute → Verify → Re-plan), multimodal document understanding (PDF/Image/Drawing), local RAG, and sandboxed tool execution. It emphasizes verified deliverables, data sovereignty, and a modular model architecture with dynamic loading, aiming to automate repetitive tasks and improve productivity within an organization's network.

KEY FINDINGS

The core innovation lies in a fully localized AI system operating on-premise, eliminating reliance on cloud APIs and ensuring data residency. The system incorporates agentic AI for autonomous task completion, supports multimodal data processing (text, PDFs, images, drawings), utilizes a model registry and dynamic loading, and includes a secure web UI for user interaction. The solution leverages quantized 7B/8B LLMs and incorporates OCR, VLM, and document processing pipelines. The architecture emphasizes data sovereignty, auditability (through logs and user roles), and a future-ready model integration strategy. A key differentiating factor is the 'Plan → Execute → Verify → Re-plan' workflow for iterative improvement.

DETAILED ANALYSIS

The proposed solution utilizes a layered architecture: a secure web UI for user interaction, an agent controller managing query understanding and task planning, and a document workflow (PDF/Scan/Text Summarization + RAG) for processing. A coding workflow enables code generation and execution within a sandboxed environment. The multimodal workflow integrates visual analysis utilizing OCR and Vision models. Model selection is driven by a task-aware selection mechanism and dynamic loading to optimize resource utilization. The core AI engine comprises quantized 7B/8B text and code LLMs alongside a vision/OCR model, dynamically loaded based on task requirements. The system prioritizes data safety through local processing, network isolation, and audit logging. Furthermore, the system provides verified deliverables, ensuring the trustworthiness of the results. The platform's architecture is designed for deployment from workstations to enterprise-level deployments, offering adaptability across different scale operations.

RISKS AND OPERATIONAL ISSUES

Potential challenges include GPU limits and contention, OCR + VLM pipeline complexity, and the potential for hallucination in the AI models. Data leakage represents a critical risk mitigated by the fully isolated environment. Complex documents present processing difficulties. Operational issues include the need for specialized technical expertise for maintenance and model updates, potentially demanding a dedicated team. The reliance on quantized models might introduce accuracy trade-offs compared to larger models. The system’s air-gapped nature introduces challenges for remote debugging or external model fine-tuning.

RECOMMENDATIONS

To mitigate identified risks, Aegis should prioritize the development of robust validation and verification mechanisms for AI outputs. Investing in thorough testing of the OCR and VLM pipelines is crucial to minimize hallucination. Continuous monitoring of GPU utilization and resource contention is recommended. A phased deployment approach, starting with controlled environments, is advised. The team should focus on streamlining the model registry and dynamic loading processes to improve efficiency. Finally, detailed audit trails and rigorous security protocols are essential for maintaining data sovereignty and operational control. Adherence to the provided PPT template is critical for consistent submission.

AUTHORITATIVE SOURCES & CITATIONS

Document Name	Referenced Pages	Relevance Status
SIH2026ppt.pdf	Page(s) 1, 2, 3, 4, 5, 6, 7	High

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